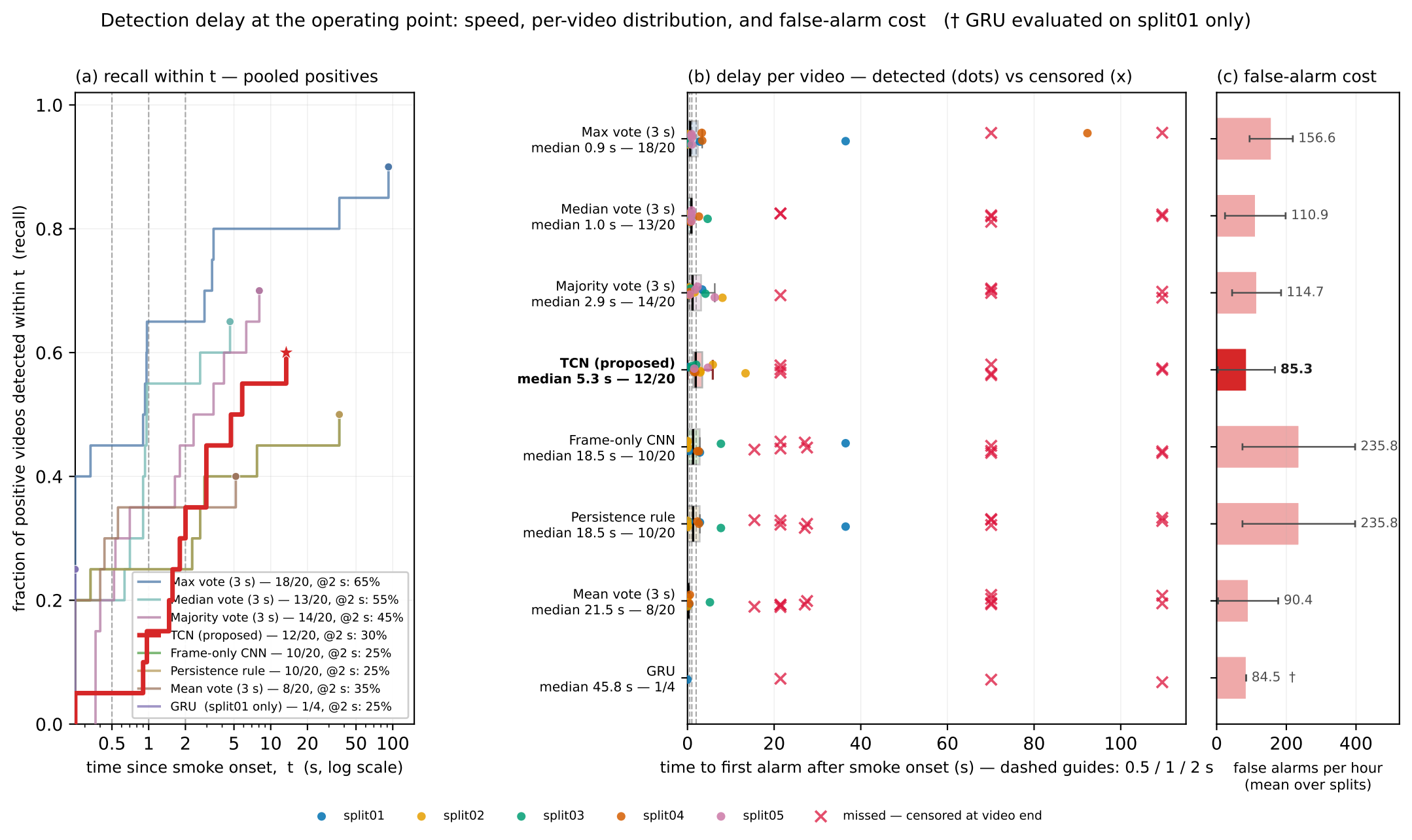
Detection delay at the operating point: speed, per-video distribution, and false-alarm cost († GRU evaluated on split01 only)
(a) recall within t — pooled positives
fraction of positive videos detected within t (recall)
time since smoke onset, t (s, log scale)
Max vote (3 s) — 18/20, @2 s: 65%
Median vote (3 s) — 13/20, @2 s: 55%
Majority vote (3 s) — 14/20, @2 s: 45%
TCN (proposed) — 12/20, @2 s: 30%
Frame-only CNN — 10/20, @2 s: 25%
Persistence rule — 10/20, @2 s: 25%
Mean vote (3 s) — 8/20, @2 s: 35%
GRU (split01 only) — 1/4, @2 s: 25%
(b) delay per video — detected (dots) vs censored (x)
Max vote (3 s)
median 0.9 s — 18/20
Median vote (3 s)
median 1.0 s — 13/20
Majority vote (3 s)
median 2.9 s — 14/20
TCN (proposed)
median 5.3 s — 12/20
Frame-only CNN
median 18.5 s — 10/20
Persistence rule
median 18.5 s — 10/20
Mean vote (3 s)
median 21.5 s — 8/20
GRU
median 45.8 s — 1/4
time to first alarm after smoke onset (s) — dashed guides: 0.5 / 1 / 2 s
(c) false-alarm cost
156.6
110.9
114.7
85.3
235.8
235.8
90.4
84.5 †
false alarms per hour
(mean over splits)
split01
split02
split03
split04
split05
missed — censored at video end